$\boxed{\text{Valid conclusion:}}$ Tofu is not tasty to eat.

f) "I am either dreaming or hallucinating." "I am not dreaming."
"If I am hallucinating, I see elephants running down the road."

$\boxed{\text{Soln:}}$ p: I am dreaming q: I am hallucinating r: I see elephants running down the road.

$\boxed{\text{Premise:}}$ 1) $p \vee q$, 2) $\neg p$ 3) $q \Rightarrow r$

Steps	Reason
1. $p \vee q$	Premise
2. $\neg p$	Premise
3. $q \Rightarrow r$	Premise
4. q	Disjunctive syllogism on 1,2
5. r	Modus ponens on 4,3

$\boxed{\text{Valid conclusion:}}$ I see elephants running down the road.

11) Show that the argument form with premises $p_1, p_2, \ldots\ldots, p_n$ and conclusion $q \rightarrow r$ is valid if the argument form with premises $p_1, p_2, \ldots\ldots, p_n, q$, and conclusion r is valid.

$\boxed{\text{Soln:}}$ Let, the argument form with premises $p_1, p_2, \ldots\ldots, p_n, q$ and conclusion r be valid. By defn. of validity, if $p_1, p_2, \ldots\ldots, p_n, q$ are all True, then r is True. $\therefore$ By defn. of AND, $(p_1 \wedge p_2 \wedge \ldots\ldots \wedge p_n \wedge q)$ is True. By defn. of r is True. $\therefore$ By defn. of $(p_1 \wedge p_2 \wedge \ldots\ldots \wedge p_n \wedge q) \Longrightarrow r$ is True. $\therefore$ By conditional-disjunction conditional, $(p_1 \wedge p_2 \wedge \ldots\ldots \wedge p_n \wedge q) \vee r$ is True. By 1st De Morgan's Law, equivalence, $\neg(p_1 \wedge p_2 \wedge \ldots\ldots \wedge p_n \wedge q) \vee r$ is True. By Associative Law, $(\neg p_1 \vee \neg p_2 \vee \ldots\ldots \vee \neg p_n \vee \neg q) \vee r$ is True. By conditional disjunction $(\neg p_1 \vee \neg p_2 \vee \ldots\ldots \vee \neg p_n) \vee (\neg q \vee r)$ is True. By de Morgan's Law, equivalence, $(\neg p_1 \vee \neg p_2 \vee \ldots\ldots \vee \neg p_n) \vee (q \Rightarrow r)$ is True. By conditional disjunction $\neg(p_1 \wedge p_2 \wedge \ldots\ldots \wedge p_n) \vee (q \Rightarrow r)$ is True. By conditional disjunction equivalence, $\boxed{(p_1 \wedge p_2 \wedge \ldots\ldots \wedge p_n) \Rightarrow (q \Rightarrow r)}$ is True.